Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Programming concepts like loops, conditionals, and drafter function will help complete my project and I expect to use most of them. I want to improve my skills in using drafters while completing this project.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the parts of converting the numbers into binary and hexadecimal will be the easiest for me because they will just require a few loops and conditionals. The part of creating the website pages will be the hardest for me because of the several pages that I would need to link for the website.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I believe building the actual website GUI will take the most amount of time and effort if I want it to look good. I think the first feature that should be implemented should be the conversion of the numbers.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Distractions that might interfere with my work could be work from other classes or text messages. I will turn my phone to Do Not Disturb while I work on the project. Resources like TA office hours are available as well as the CISC discord server while working on the project.

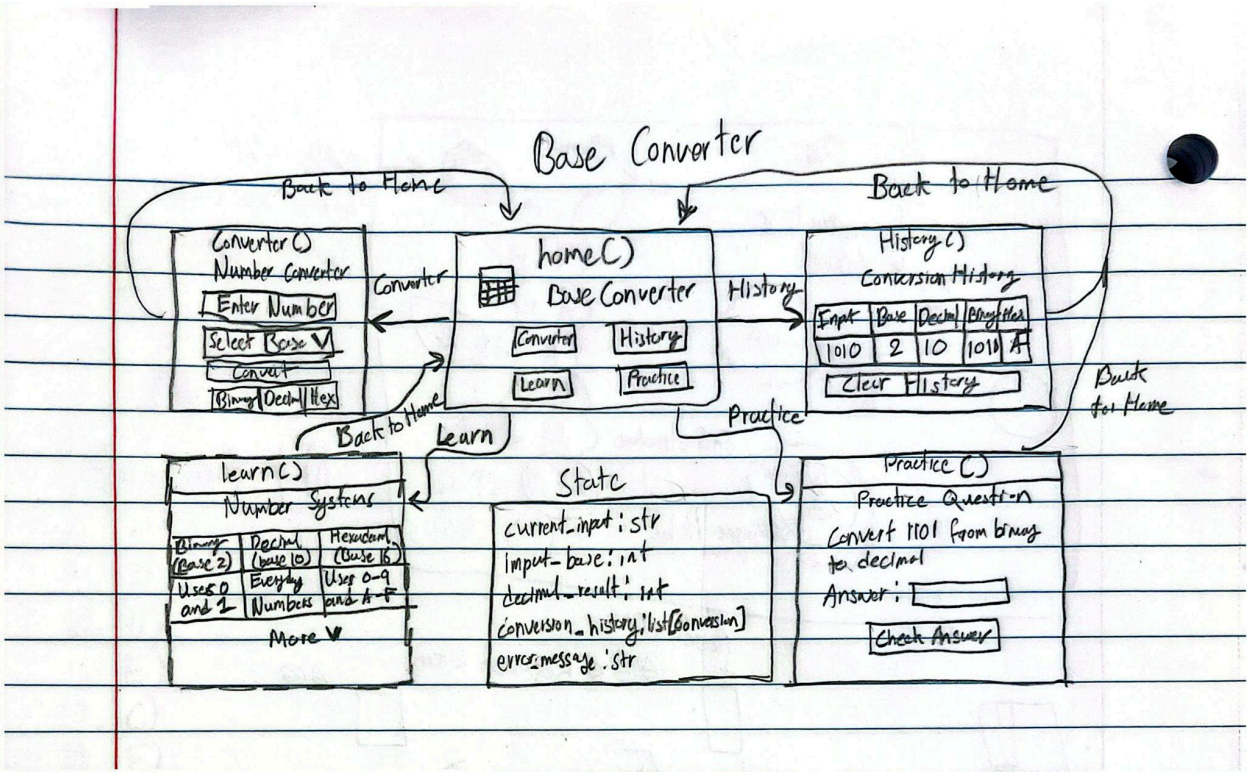
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Base Converter
Back to Home
Back to Home
converter()
Number Converter
Enter Number
Select Base
Convert
Binary
Decimal
Hex
Converter
home()
Base Converter
Converter
History
Learn
Practice
History
History()
Conversion History
Input
Base
Decimal
Binary
Hex
1010
2
10
1010
A
Clear History
Back to Home
Back to Home
Learn
Practice
learn()
Number Systems
Binary (Base 2)
Decimal (base 10)
Hexadecimal (Base 16)
Uses 0 and 1
Everyday Numbers
Uses 0-9 and A-F
More
State
current_input: str
input_base: int
decimal_result: int
conversion_history: list[Conversion]
error_message: str
Practice()
Practice Question
Convert 1101 from binary to decimal
Answer:
Check Answer

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**The purpose of the site is to convert numbers into binary and hexadecimal while also teaching the user how the conversion works.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	Current_input - str - Stores numbers entered by a user and updates when the user enters new numbers. input _base - int - Stores a number system then updates when user selects a new conversion Error_message - str - Stores an error message when the user inputs an invalid number. Clears when the conversion was successful. Decimal_result - int - Stores the decimal equivalent of the user's input and updates when the conversion works. Conversion_history - list[Conversion] - Stores previous conversions as Conversion dataclass. It is updated when the user completes a new conversion.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified	class Conversion: original_number - str - number entered by the user. Decimal_system - int - Decimal number of the

(if they are modified)	equivalent input. Binary_value - int - Binary value of the user input. Hexadecimal_value - int - Hexadecimal value of the user input.
Routes (minimum 7) The names of your route functions	home converter convert_number history clear_history learn practice check_answer
Pages (minimum 5) A brief description of the pages in your site	Home Page - Gives a description of what the website does and also includes navigation to other pages. Converter Page - Allows people to enter in a number and a number system to convert it into. History Page - Shows the history of all the calculations that have been done. Learning Page - Shows a description of how to

	convert each number into the respective number system. Practice Page - The user can practice trying to convert numbers into other number systems.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	Number entry - TextBox - User inputs a number that they want to convert Original system - SelectBox - The number system that the user put in. Practice Answer - TextBox - The answer that the user inputs for the practice
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	Input checker - Checks whether the input is valid in the selected system. Conversion selection - Checks what system the user selected to convert Practice_answer_check - Checks whether the practice answer matches the correct answer. History_is_empty - Checks whether the history list is empty.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	Using a for loop over each of the items in the conversion_history so that it shows all the conversions in the list at once rather than opening a new page for each conversion.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Didn't use LLM

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.